

PROJECT DOCUMENTATION

Metro Ticket Generating System in ServiceNow

1. Introduction

1.1 Project Overview

Project Title: Metro Ticket Generating System in ServiceNow

The project digitizes metro ticket booking using ServiceNow. It provides a Service Catalog booking interface, automated fare calculation, payment selection, QR-based digital ticket presentation, and automated storage of ticket information.

1.2 Purpose

The purpose is to reduce manual ticketing effort, improve fare accuracy, support digital payment choices, provide a digital QR ticket, and maintain consistent operational records.

2. Project Overview

Key features:

- Metro Station's Details master table.
- Book A Metro Ticket Service Catalog item.
- Starting From and Going To station references.
- Single and Return journey support.
- 1–4 passenger selection.
- Automatic fare calculation.
- UPI, Card and Others payment options.
- Conditional Enter Payment Mode field.
- Metro QR Widget for digital ticket presentation.
- Metro Database with MTKT auto-numbered ticket records.
- RITM Number tracking.
- Flow Designer automation.
- ServiceNow ACL-based access control.

3. Architecture

Frontend / User Experience: ServiceNow Service Portal and Service Catalog.

Application Logic: Catalog Client Script for fare/validation, Catalog UI Policy for conditional fields, and Flow Designer for automation.

Database: ServiceNow custom tables u_metro_station_details and u_metro_database.

4. Setup Instructions

Prerequisites: Access to a ServiceNow development instance with permissions to create tables, catalog items, scripts, widgets, flows and ACLs.

- Create Metro Station's Details table: `u_metro_station_details`.
- Create station master records.
- Create Book A Metro Ticket catalog item and nine variables.
- Configure the Fare Client Script.
- Configure the Others payment Catalog UI Policy.
- Create Metro QR Widget and QR generation Catalog Client Script.
- Create Metro Database table and required fields.
- Create and activate Metro Project Flow.
- Associate the flow with Book A Metro Ticket.
- Create/select the final Update Set for deployment.

5. Configuration Structure

Area	Configuration
Master Data	Metro Station's Details — <code>u_metro_station_details</code>
Catalog	Book A Metro Ticket
Variables	Starting From, Going To, Type of journey, No of Passengers, Amount for single journey, Amount including return, Mode of Payment, Enter payment mode, QR Generated
Client Script	Fare Client Script
UI Policy	Others payment conditional visibility
Widget	Metro QR Widget
Transaction Table	Metro Database — <code>u_metro_database</code>
Automation	Metro Project Flow
Security	Custom table ACLs / roles
Numbering	MTKT + 5 digits

6. Running the Application

- Open the ServiceNow Service Portal.
- Open Book A Metro Ticket.
- Select Starting From and Going To.
- Choose journey type and passenger count.
- Verify the calculated amount.
- Select UPI, Card or Others; if Others, enter the payment mode.
- Complete the booking/QR flow.
- Verify the request/RITM.
- Open Metro Database and verify the generated ticket record.

7. API / Integration Documentation

Integration	Purpose	Input	Output
QRServer API	Generate QR image representation.	Encoded ticket URL/identifier.	QR image URL.
ServiceNow Flow Designer	Automate catalog-to-database record creation.	Requested Item + catalog variables.	Metro Database record.

8. Authentication & Authorization

Authentication is provided by the ServiceNow platform. Authorization for the custom station table uses ServiceNow ACL/role controls. The custom table's read access was configured with the generated user role.

9. User Interface

The main user interface is the ServiceNow Service Portal booking form. It contains station, journey, passenger, fare, payment and QR-related fields.

10. Testing

Testing covers positive and negative catalog scenarios, fare calculation, conditional payment fields, QR display, Flow Designer record creation, RITM mapping and access control.

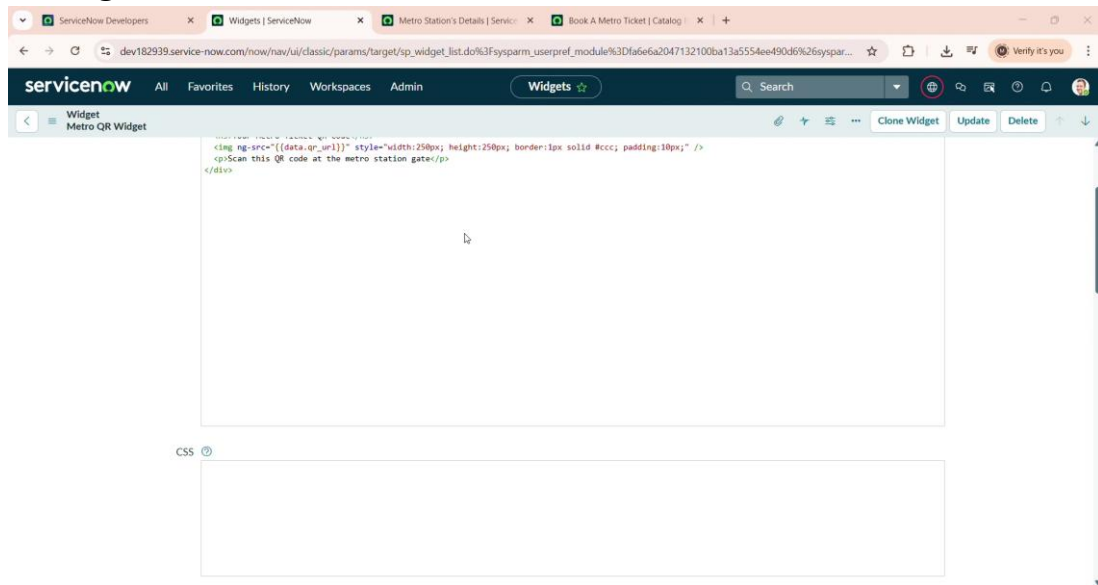
11. Screenshots / Demo Evidence

Catalog item / configuration

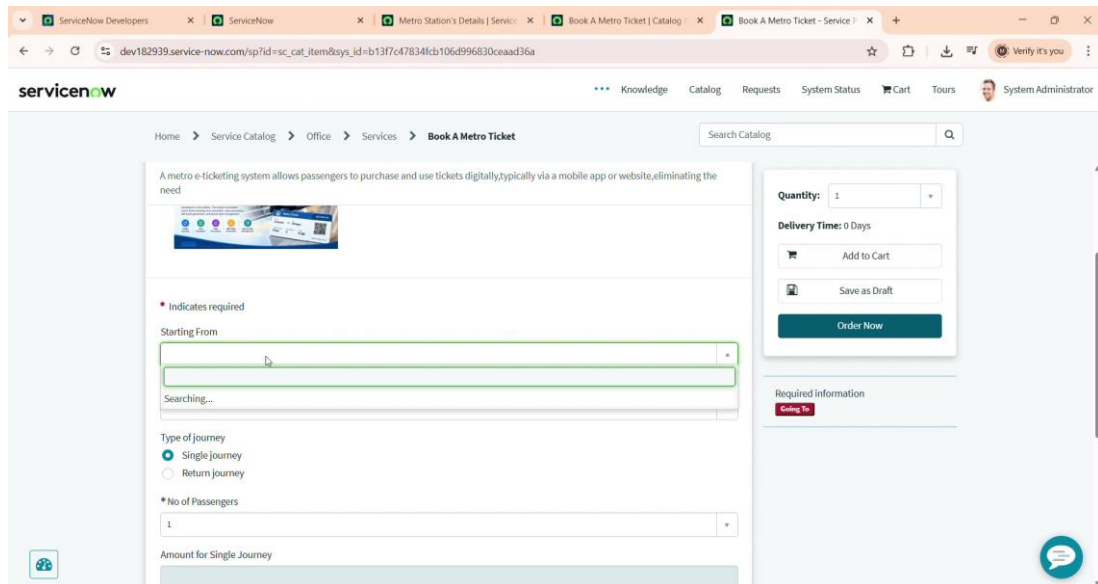
The screenshot shows the ServiceNow Catalog Item configuration page for 'Book A Metro Ticket'. The page includes a header with navigation links and a main content area with a table of assigned topics. The table has columns for Name, Active, Table, View, Type, and Updated. The 'QR Generation' topic is highlighted.

Name	Active	Table	View	Type	Updated
Calculate Fare - Starting From	true			onChange	2026-09-08 23:17:44
Calculate Fare - Going To	true			onChange	2026-09-08 23:19:02
Fare auto calculation	true			onChange	2026-09-08 23:07:52
Calculate Fare - Journey Type	true			onChange	2026-09-08 23:20:57
QR Generation	true			onSubmit	2026-09-08 22:02:29

QR widget configuration



Metro ticket form



Completed booking / fare

ServiceNow DevelopersServiceNowMetro Station's Details | ServiceBook A Metro Ticket | CatalogBook A Metro Ticket - Service

dev182939.service-now.com/sp?id=sc_cat_item&sys_id=b137c47834fcb106d996830ceaad36a

Verify it's you

KnowledgeCatalogRequestsSystem StatusCartToursSystem Administrator

Home > Service Catalog > Office > Services > Book A Metro Ticket

Search Catalog

Return Journey

* No of Passengers

2

Amount for Single Journey

Amount Including Return

200

Mode of Payment

Card

Others

UPI

* Enter Payment Mode

Add attachments

Quantity: 1

Delivery Time: 0 Days

Add to Cart

Save as Draft

Order Now

Required information

Enter Payment Mode

Request summary

ServiceNow DevelopersServiceNowMetro Station's Details | ServiceBook A Metro Ticket | CatalogRequest Summary - REQ0010006

dev182939.service-now.com/sp?id=sc_request&is_new_order&table=sc_request&sys_id=514cfd5839f4f506d996830ceaad35c

Verify it's you

KnowledgeCatalogRequestsSystem StatusCartToursSystem Administrator

Home > Request Summary - REQ0010006

Search Catalog

Submitted : 2026-09-12 01:51:42
Request Number : REQ0010006
Estimated Delivery : 2026-09-12

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Book A Metro Ticket	2026-09-12		---	1	---
					Total: £0.00

Metro Database

Ticket Number	RITM Number	Starting From	Going To	No of Passengers	Type of Journey	Amount for Single Journey	Amount Including Return	Mode of Payment	Enter
MTKT00003	RITM0010006	Ameerpet	LB Nagar	2	2		200	others	cash
MTKT00002	RITM0010005	Jubilee Hills	Panjabgutta	1	2		40	card	

12. Known Issues

- QR Generated can remain false if the QR generation step is not triggered during a test.
- The ServiceNow Request Item field is intentionally retained as a reference; the readable RITM Number is stored separately.
- QR image generation depends on the availability of the configured external QR service.

13. Future Enhancements

- Real payment gateway integration.
- Production metro gate QR validation.
- Dynamic fare master table instead of a hard-coded fare matrix.
- Passenger notifications through email/SMS.
- Admin dashboards and usage analytics.
- More stations, zones and fare rules.
- Mobile-optimized ticket presentation.
- Deployment through controlled DEV → TEST → PROD update-set process.

Project Report Format

The final report is organized according to the supplied report format:

Section	Covered By
1. Introduction	Project Overview and Purpose
2. Ideation Phase	Ideation Phase consolidated document
3. Requirement Analysis	Requirement Analysis consolidated document
4. Project Design	Project Design Phase consolidated document
5. Project Planning & Scheduling	Project Planning Phase consolidated document
6. Functional and Performance Testing	Project Development Phase consolidated document

7. Results	Demo screenshots and test outcomes
8. Advantages & Disadvantages	Summary below
9. Conclusion	Summary below
10. Future Scope	Future Enhancements
11. Appendix	Configuration, source code references and deployment notes

Advantages & Disadvantages

Advantages	Disadvantages / Limitations
Automated fare calculation	Fare matrix is currently configured in client-side code.
Digital QR ticket representation	External QR service dependency.
Centralized ServiceNow records	Production payment integration is not included.
Multiple payment choices	Current payment entry is informational rather than a live gateway.
Reduced manual effort	Full metro gate validation is outside the current scope.
Readable ticket/RITM tracking	Availability depends on ServiceNow platform.

Conclusion

The Metro Ticket Generating System demonstrates how ServiceNow can be configured as a digital metro ticketing workflow. The solution connects a Service Catalog experience with automated fare logic, QR presentation, Flow Designer automation, custom data tables and operational tracking.

Appendix

Item	Details
Key Table 1	u_metro_station_details — Metro Station's Details
Key Table 2	u_metro_database — Metro Database
Catalog Item	Book A Metro Ticket
Flow	Metro Project
QR Widget	metro_qr_widget / Metro QR Widget
Ticket Number	MTKT + 5 digits
RITM Tracking	RITM Number mapped from Requested Item → Number
Deployment	ServiceNow Update Set: Metro Ticket Generating System - V1.0